CLASS 8 – CBSE CHEMISTRY QUESTION PAPER

Time Allowed: 1 Hour

Maximum Marks: 30

SECTION A: SHORT ANSWER QUESTIONS

(4 questions × 2 marks = 8 Marks)

Q1. Write balanced word equations for the following chemical reactions:

(a) Magnesium reacts with Oxygen to form Magnesium Oxide.

(b) Iron reacts with Copper Sulphate solution to form Iron Sulphate and Copper.

Q2. Differentiate between thermoplastic and thermosetting plastics based on their structure and behavior upon heating. Give one example of each.

Q3. What is meant by the calorific value of a fuel? State its SI unit.

Q4. Define displacement reaction. What happens when a copper strip is dipped into a Zinc Sulphate solution? Give a reason for your answer.

SECTION B: MEDIUM ANSWER QUESTIONS & EQUATIONS

(4 questions × 3 marks = 12 Marks)

Q5. Chemical Equations & Reactions:

Complete and balance the following chemical equations:

(a) $Na + H_2O \rightarrow _____ + H_2 \uparrow$

(b) $SO_2 + H_2O \rightarrow _____$

(c) $Zn + H_2SO_4 \rightarrow _____ + H_2 \uparrow$

Q6. Fossil Fuels & Conservation:

Explain the process of destructive distillation of coal. Name two useful products obtained from this process along with their primary industrial uses.

Q7. Metals & Non-Metals:

(a) How do metals and non-metals differ in their reaction with acids? *(2 marks)*

(b) Which gas is produced when a metal reacts with a dilute acid? How do you test for the presence of this gas? *(1 mark)*

Q8. Flame & Combustion:

Draw a neat labeled diagram showing the three distinct zones of a candle flame. Identify which zone has the highest temperature and explain why.

SECTION C: LONG ANSWER & DETAILED QUESTIONS

(2 questions × 5 marks = 10 Marks)

Q9. Corrosion & Chemical Properties:

(a) What is rusting of iron? Write the chemical equation for the rusting process. *(2 marks)*

(b) Explain why copper vessels develop a greenish coating when exposed to moist air over time. Write the chemical equation representing this reaction. *(2 marks)*

(c) List two methods used to prevent the corrosion of metals. *(1 mark)*

Q10. Polymerization & Environmental Impact:

(a) Why are plastics widely used in daily life despite their harmful impact on the environment? Mention three desirable properties of plastics. *(2 marks)*

(b) Differentiate between biodegradable and non-biodegradable substances with suitable examples. *(2 marks)*

(c) State the "5R Principle" recommended for reducing the environmental hazards caused by plastic waste. *(1 mark)*